

Flood Polynomial Model (MKM-FPO-001)

Wolfram Oracle Benchmark of v1.0 and its GBM Approximation

MKM Research Labs

Version 0.1 — 03-November-2026

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1 Executive Summary

This document benchmarks the Flood Polynomial Model (MKM-FPO-001), v1.0, a closed-form approximation of the per-gauge GBM flood classifiers (MKM-FC-001) that the single-gauge stress scenario uses to price PRS trades hour by hour before a breach. It asks whether the code computes what its model document (`flood_poly.pdf`, v1.0) says, whether the document's claims hold, and how well the polynomial approximates the classifiers it stands in for.

The code computes its documented formula. `p_flood_poly` agrees with a 50-digit implementation of Eqs. 1–4 and the printed coefficients, and so do `p_flood_series`, the copy in `stress.sensitivity` and its slope. The port stress screen's JavaScript holds the same coefficients (Table 1). Wolfram, from the exact coefficients, agrees with the reference to 3.6×10^{-31}.

But the document's evidence does not hold (Table 3).

- **Its tables have no source.** The code reproduces 8 of the 54 cells of Table 1 and none of Table 2. Nor does the GBM reproduce them. No generator for either table ever existed.
- **At the severe level, P does not rise from 0.42 to 0.97.** It is 0.00 at hour 0 and 0.29 at hour 167, after peaking near 0.97 around hour 96.
- **The fit is not as stated.** On every real storm state of the fitted gauge, R^2 is 0.908, not 0.9395. MAE is 0.0103, better than the stated 0.0456. The claimed 0.05 accuracy is not met near severe.

The approximation, measured. The comparison uses the 11 gauges' classifiers on the real storm states they are trained on (Table 2).

- Below severe, the polynomial is within 0.0078 to 0.0228 of the GBM on average.
- Near severe (75–100%), where stress P&L is decided, the mean error is 0.052 to 0.073, and 15 to 27% of states are off by more than 0.10.
- The bias near severe ranges from -0.020 to +0.048. It is right on average but imprecise state by state.
- Early in a storm the surface is not monotone in the level: at hour 24, P passes $\frac{1}{2}$ twice below severe and falls as the water rises between the two levels.

Two things production does are not in the document. The stress route latches $P = 1$ once the level reaches severe. And `p_flood_simple`, which sets portfolio stress P&L, is the ratio $\min(1, w/s)$, not a probability.

Evidence. 8 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `6a41a963dd7a`.
- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-11-03 by `python -m tools.wolfram_oracle fpo001`.

2 Model Purpose and Scope

2.1 What is benchmarked

- `src/models/stress/flood_poly.py`: `p_flood_poly`, `p_flood_series` and `p_flood_simple`.
- The copy in `src/models/stress/sensitivity.py`.
- The coefficients hard-coded in the port stress screen.
- The polynomial against the production GBM classifier of every gauge that has one.

2.2 What is not

- The classifiers themselves (MKM-FC-001).
- The stress scenarios' hydrographs.
- The P&L built on P.

This is a benchmark of the implementation and of the approximation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

The document's model is

$$P = \sigma(ah + bt + cht + dh^2 + et^2 + f), \quad h = \ln \max(w/s, 10^{-8}), \quad t = \ln \frac{\text{hour} + 1}{168}, \tag{1}$$

with the exponent clamped to ± 30. The reference evaluates it at 50 digits from the printed coefficients.

What the polynomial can know. Each gauge's GBM also sees the rate of rise Δh and its change. A function of (h, t) alone can at best reproduce the GBM's average over the storm states that share a level and an hour. The approximation is therefore judged on those real states, as production meets them, not on a grid at $\Delta h = 0$. A grid includes states storms do not produce, and states storms do produce are left out of it.

4 Benchmark Method

The code. The code is checked against the reference over three severe thresholds, sixteen level ratios from zero to 1.5, and thirteen hours.

Wolfram. From exact coefficients, Wolfram evaluates the clamped logistic at every cell of Table 1, the levels where $P = \frac{1}{2}$ (by `NSolve` over the reals below severe), and $\partial P/\partial h$ symbolically.

The GBM comparison.

- For every gauge with a classifier, 2,000 storms are drawn once (seed 7) from the catalogue. Each storm's states are rebuilt through production's training-vector builder.
- The polynomial and the GBM are compared on the states below severe.
- For the fitted gauge, the fit metrics are recomputed on every state of the full catalogue. The cubic of §9.1 is fitted to the same states.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Limit	Outcome
Eqs. 1–4: `p_flood_poly` against the reference, worst relative error	1.8×10^{-14}	1.0×10^{-12}	pass
A threshold at or below zero gives zero	yes		pass
`p_flood_series` is `p_flood_poly` hour by hour	0	0	pass
§5.2: `p_flood_simple` is $\min(1, w/s)$	6.9×10^{-17}	1.0×10^{-12}	pass
The copy in `stress.sensitivity`, worst relative error	1.8×10^{-14}	1.0×10^{-12}	pass
Its closed-form $\partial P/\partial h$, worst relative error	1.1×10^{-13}	1.0×10^{-12}	pass
The port stress screen's JavaScript coefficients are the document's	yes		pass
Wolfram agrees with the reference (Table 1 cells, $P = \frac{1}{2}$ levels, slopes)	3.6×10^{-31}	1.0×10^{-25}	pass

5.1 The gauges

Table 2: The polynomial against each gauge's GBM on real storm states, near severe ($w/s \geq 0.75$) and below it

Gauge	States	GBM	Poly	Mean $\lvert d\rvert$	95%	> 0.1	All below
GAUGE-b2dbe58b	14,592	0.326	0.367	0.071	0.249	26%	0.0150
GAUGE-dc5e88fb	9,894	0.296	0.305	0.052	0.186	15%	0.0098
GAUGE-1aaf8b08	18,815	0.413	0.436	0.061	0.210	21%	0.0228
GAUGE-d208bf3d	14,990	0.445	0.424	0.061	0.207	19%	0.0225
GAUGE-c0fbc443	10,256	0.259	0.295	0.061	0.233	21%	0.0096
GAUGE-a6be4ee9	11,406	0.300	0.328	0.063	0.226	22%	0.0111
GAUGE-ef53398c	8,379	0.281	0.291	0.052	0.209	17%	0.0078
GAUGE-085522ee	16,268	0.416	0.421	0.060	0.203	19%	0.0186
GAUGE-9fffb377	12,730	0.287	0.335	0.073	0.264	27%	0.0119
GAUGE-f8629444	14,105	0.406	0.395	0.059	0.197	18%	0.0182
GAUGE-9e5f3ef9	17,404	0.374	0.407	0.068	0.244	24%	0.0199

5.2 The model document's claims

Table 3: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§8, Table 1	P by level and hour, $s = 6.58$ m	8 of 54 cells reproduced by the code	**not met**
§8, Table 2	GBM and polynomial at six points	polynomial column 0 of 6; GBM column 0 of 6 (rate of rise zero)	**not met**
§8	at the severe level, 0.42 at hour 0 rising to 0.97 at hour 167	0.00 at hour 0, 0.29 at hour 167	**not met**
§8	$\lvert\text{poly} - \text{GBM}\rvert < 0.05$ across the surface	near severe, 95th percentile up to 0.26; above 0.05 in up to 39% of states	**not met**
§4.2	R^2 0.9395, MAE 0.0456, RMSE 0.0986 against the GBM	0.9083, 0.0103, 0.0444 on all 1,680,000 real states	**not met**
§9.1	a cubic kernel reaches $R^2 \approx 0.97$	0.9460 on the same states	**not met**
§4.2	mildly conservative: over-predicts near severe	bias near severe -0.020 to +0.048 across gauges	**not met**
§3.2	$t \approx -5.1$ at hour 0	-5.1240	met
—	rising water raises P	at hour 24, P passes $\frac{1}{2}$ at 52% and 74% of severe: it falls as the water rises between them	finding
§5	the scenario uses P as given	the stress route latches $P = 1$ once the level reaches severe; undocumented	finding
§5.2	`p_flood_simple` for portfolio stress	$\min(1, w/s)$ is a ratio, not a probability, and it sets portfolio stress P&L	finding

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The GBMs are the reference here; whether they are right is MKM-FC-001's question.

2. **Remediation.** The coefficients are near the best this form can do on real states: refitting the same six terms does not reduce the error near severe. The model is refitted as v2.0, pooled across gauges, with a richer kernel constrained to rise with the level and $P = 1$ at or above severe. Its document is reissued with tables generated from the code, and it is benchmarked as v0.2 of this document.

7 Document History

Date	Version	Description	Author
03-Nov-2026	0.1	First issue: tests v1.0	David K Kelly